

CLINISCAN: LUNG ABNORMALITY DETECTION ON CHEST X-RAYS USING AI

Milestone 1: Data Preparation & Initial Setup

1. Introduction

Medical imaging plays a critical role in the diagnosis of pulmonary diseases. Chest X-rays (CXR) are one of the most commonly used imaging modalities due to their low cost, speed, and wide availability. Manual interpretation of X-ray images is time-consuming and highly dependent on the expertise of radiologists.

To address these challenges, this project utilizes a large-scale, annotated chest X-ray dataset to develop an AI-based lung abnormality detection system.

2. Dataset Description (Theory)

The project uses the VinBigData Chest X-ray Abnormalities Detection dataset, which contains 15,000 chest X-ray images annotated by professional radiologists. The dataset includes bounding box annotations for multiple lung and cardiac abnormalities.

Images are provided in DICOM format, which is commonly used in medical imaging but not directly suitable for deep learning frameworks. Therefore, preprocessing is essential to convert and standardize the data.

3. Data Preprocessing

Medical images differ significantly from natural images due to:

- High resolution
- Device-dependent intensity ranges
- Presence of noise
- Non-uniform contrast

Preprocessing ensures:

- Faster model convergence

- Improved generalization
- Reduced computational cost
- Better clinical interpretability

4. Data Preprocessing Steps

4.1 DICOM to PNG Conversion

DICOM images are converted to PNG format to make them compatible with computer vision libraries such as OpenCV and deep learning frameworks.

4.2 Grayscale Conversion

Chest X-ray images are inherently grayscale. Explicit grayscale conversion ensures consistency across the dataset.

4.3 Image Resizing

All images are resized to a fixed resolution (e.g., 512×512 pixels) to maintain uniform input dimensions for neural networks.

4.4 Intensity Normalization

Min-Max normalization is applied to scale pixel values between 0 and 255. This stabilizes training and prevents dominance of extreme intensity values.

4.5 Contrast Enhancement (CLAHE)

Contrast Limited Adaptive Histogram Equalization (CLAHE) improves visibility of lung structures while preventing over-amplification of noise.

4.6 Noise Reduction

Gaussian blurring is applied to reduce high-frequency noise while preserving anatomical edges.

4.7 Border Cropping

Unnecessary black borders are removed to focus the model on the anatomical region of interest.

4.8 Padding

Padding ensures that all images maintain consistent height and width after cropping operations.

4.9 Annotation Conversion

Annotations are converted from CSV format to YOLO format, which represents bounding boxes using normalized center coordinates and dimensions.

4.10 Data Augmentation

Safe augmentations such as horizontal flipping, small rotations, mild zoom, and contrast adjustment are applied to improve robustness without altering clinical meaning.

4.11 Train-Validation-Test Split

The dataset is split into training, validation, and testing sets to ensure unbiased performance evaluation.

5. Outcome of Milestone-1

- Dataset is verified and structured
- Images are converted and preprocessed
- Annotations are model-ready
- The pipeline is scalable for full-dataset training